Mancuna AI

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Problem Analysis

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Table 1: Starting Mancuna board

- Board is 2x2.
- Pieces move counter-clockwise.
- There is a total of 8 pieces.
- Tie repetition length of 3.
- Maximum of two possible moves each turn.
- If there are no legal moves, the opponent wins by default.
- The score is based on how many pieces are in each player's board.

Problem Analysis

When looking at the search space abstractly we can infer certain information.

- There are about ~ 150 possible board combinations.
- Considering who makes the next move, there are about ~ 300 possible board combinations

Problem Analysis

Looking at the search space during all possible match combinations, some more interesting conclusions can be drawn.

During a match there are only **113 possible states**, taking into account who makes the next move.

With a branching factor of 2 at most, the search space **seems extremely small**.

Problem Analysis

In reality such is not the case.

The tie repetition rule makes it so that each branch in the search space is different: **caching is not possible**.

Because caching is not possible, and given that there are tie loops of over 30 plays long, the search space quickly becomes over 2^{30} which **exceeds a billion different branches**.

Best Option AI

Best Option AI

The idea is to implement a minmax-like algorithm while disregarding the score and looking only at **the worst-case outcome** for a play.

When looking at the search space in this manner, **caching becomes possible again**.

This means that the minmax must **at most calculate 113 states**, at which point it already knows the entire search space.

Best Option AI

Upon having the worst-case outcome for every single possible board, the AI executes the following heuristic.

- If there is only one legal move, it will play it blindly.
- If one of the moves has the worst-case outcome of a win, that move will be played.
- If no winning moves exist, the first move with the worst-case outcome as a tie will be played.
- If there are no winning or tying moves, the first move is played.

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Results

On both sides of the board this AI ties with the expert AI, achieving a score of 5-3 regardless of going first or second.

On the first move for both sides of the board, this AI has at least one move which has a worst-case outcome of a tie.

This means that this is a **complete game** where only ties are possible when both players play optimally.

Best Score AI

Best Score AI

As explained eariler, we cannot do a blind search across the entire search space.

However, we can **exploit the Best Option AI** to search only through a very small portion of the entire search space.

Best Score AI

Whenever the Best Option AI is in a situation where both options lead to a tie, we can execute minmax through the search space considering only the tying moves.

With this, we get a **worst-case tying score** for each move.

Upon having this score, this AI behaves exactly like the Best Option AI, however, when both options are ties, it will choose the one that is guaranteed a higher score.

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Results

When playing **first**, it ties **4-4** with the expert AI.
When playing **second**, it ties **5-3** with the expert AI.

Best Score AI

When playing optimally, the **first** player can only tie with a score of **at most 3**. While the **second** player can only tie with a score of **at most 5**.

With this, we can conclude that **the game is solved** and the **Best Score AI plays optimally**.

If it plays optimally, why does it get a lower score against the expert AI than the "dumber" Best Option AI?

Best Score AI

The reason for the "smarter" AI to perform worst is that it **assumes both players play optimally**.

The expert AI does not play optimally. And as such it has patterns that can be exploited.

By cheer luck, the Best Option AI, as it plays double tying moves blindly, plays a pattern in which the expert AI **makes mistakes** and ends up tying with a higher score.

As the Best Score AI only plays optimally, and by **cheer luck** in the optimal play the expert AI **does not make the same mistakes**, it ends up with a lower score. Even then the expert AI makes one mistake, which leads the Best Score AI to be able to achieve a score of 4-4 instead of the expected 3-5.

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Final AI and remarks

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The final AI used was the **Best Option AI**. Although worse, it performs better for the competition.

For playing optimally the Best Score AI would be used, as it plays optimally.

For the competition, an AI could be developed that specifically exploits the expert AI. With that strategy it could be **theoretically possible to achieve higher scores or even win against the expert AI**, depending on if it has any path of play in which it makes worse blunders.

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